Tab 1

80% of n8n

PART 1 — FOUNDATION

- What is n8n & Why n8n? (vs Zapier, Make, pabbly)
- History of Automations *(brief)*
- 3 Types of Workflows in n8n *(Standard Workflow, AI-Enhanced Workflow, AI Agentic Workflow)*
- n8n Self-host vs n8n.io

PART 2 — INTERFACE & CORE CONCEPTS

- Interface Walkthrough
- Triggers → Webhook vs Cron
- Nodes & Actions
- Credentials & Authentication
- Test vs Production
- Save & Execution History & Pinning Data

PART 3 — DATA FUNDAMENTALS

- Understanding JSON
- Fixed vs Expression
- `$json` syntax
- Set Node & Data Mapping

PART 4 — PROJECT 1 — Lead Capture via Webhook *Type — Standard Workflow*

- Webhook Trigger → Capture form data from website
- Code Node → Filter genuine vs fake leads
- Store leads
- Error Handling basics

PART 5 — PROJECT 2 — Daily Lead Report via Cron → Slack *Type — Standard Workflow*

- Cron Trigger → Fetch leads datewise
- Code Node → Count, group, filter
- Send formatted summary to Slack

PART 6 — PROJECT 3 — Lead Qualifying + Personalized Email *Type — AI-Enhanced Workflow*

- Introduce AI-Enhanced Workflows here *(AI does tasks inside a fixed flow — you still control the path)*
- Webhook captures lead + their questionnaire answers from website
- AI reads responses → qualifies lead *(genuine interest vs noise)*
- AI writes personalized email based on their selections → sends automatically
- Mental Health Report angle — User answers questions → AI analyzes → generates personalized report → sends to email
- This teaches: AI node, Prompt Engineering inside n8n, passing `$json` data into AI prompts dynamically

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PART 7 — PROJECT 4 — Trained Website Chatbot *Type — AI Agentic Workflow (Single Agent pattern)*

- Introduce AI Agentic Workflows *(AI now controls the flow — not just doing a task inside it)*
- Configure chatbot widget on website
- Train it on custom data *(your use case, FAQs, product info)* via Google Doc. or Sheets
- Connect memory so it remembers conversation context
- Tools the agent can use *(search knowledge base, fetch data, trigger other workflows)*
- This teaches: Chat Trigger node, Memory, Tools

PART 8 — WRAP UP & WHAT'S NEXT

- Recap of all 3 types through the 4 projects

Project	Type	Core Skill
Lead Capture	Standard Workflow	Webhook, Code Node
Lead Report	Standard Workflow	Cron, Slack
Summarize Health Report + Email	AI-Enhanced Workflow	AI node, Prompting
Website Chatbot	AI Agentic Workflow	LLM, Memory, Tools,

- Multi-Agent pattern

Tab 2

1. What is n8n?

n8n is an **open-source, node-based workflow automation tool** that lets you connect apps, automate tasks, and build AI-powered workflows — without writing much code.

Concept > Any Tool

Simple Scenario:

> You run a business. Someone fills a form on your website. You want their data saved in a spreadsheet, a Slack message sent to your team, a welcome email sent to them, and a lead score calculated — all automatically, the moment they submit. That's exactly what n8n does.

Who is it for?

User Type	How they use n8n
Developers	Build complex automations with custom code and APIs
Business owners	Automate repetitive tasks without a dev team
Marketers	Automate lead capture, email sequences, reporting
AI builders	Build AI agents and agentic workflows visually

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2. Why n8n? (vs Zapier, Make, Pabbly)

Quick Positioning

> Zapier for simple connections. Make for visual power users. Pabbly for budget-conscious teams. **n8n when you need full control, AI, and don't want to pay per task.**

Full Comparison Table

Feature	n8n	Zapier	Make	Pabbly
Open Source	✅	❌	❌	❌
Self-hostable	✅	❌	❌	❌

Native Integrations	~1,000+	6,000+	1,500+	~1,000+
Connects to any API?	✅ HTTP Node	⚠️ Limited	⚠️ Limited	⚠️ Limited
Custom Code (JS)	✅ Full	❌	⚠️ Limited	❌
AI Agents native	✅ ~70 AI nodes, LangChain native	⚠️ Basic	⚠️ Basic	❌
Pricing model	Per workflow execution	Per task — punishes complexity	Per operation	Flat rate
Best for	Power users & devs	Non-technical beginners	Visual thinkers	Budget-friendly SMBs
Data privacy	✅ Full (self-host)	❌ Their servers	❌ Their servers	❌ Their servers
Learning curve	Medium-High	Easy	Medium	Easy

When to Pick Which — Decision Table

Situation	Pick this
Just starting, want quick wins	Zapier
Complex visual flows, good budget	Make
Tight budget, basic automations	Pabbly
AI agents, custom logic, data privacy, scale	**n8n**

Scenario that makes the choice obvious:

> You want to capture leads, score them with custom logic, qualify them using AI, and send personalized emails — all in one workflow. Zapier hits its limits immediately. Make gets expensive. Pabbly can't handle the AI layer. n8n does all of it, and you can self-host it so your lead data never leaves your server.

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3. History of Automations

Humans have always tried to do more with less effort. Automation didn't start with computers — it started with a spear.

Era	Year Range	What Humans Automated	How
Early Tools	1,000,000 BC+	Hunting & cutting	Made spears, axes — tools that multiplied human force
Water Management	3000 BC	Transporting & lifting water	Canals, aqueducts, shaduf (lever system to lift water)
Wind & Water Power	200 BC	Grinding grain, pumping water	Waterwheels and windmills — nature doing the work
Mechanical Clocks	1300s AD	Tracking time	Gears and springs replacing human timekeeping
Printing Press	1440s	Copying documents	Gutenberg's press — one person printing thousands of pages
Industrial Revolution	1760s–1840s	Manufacturing	Steam engine, automated looms — machines replacing manual labor at scale
Assembly Line	1900s	Mass production	Ford's moving assembly line — same task, repeated perfectly, at speed
Early Computers	1940s–1950s	Complex calculations	Machines doing math faster than any human
Business Software	1970s–1980s	Data entry, accounting	Spreadsheets (Lotus, Excel) — automating manual record keeping
Internet & APIs	1990s–2000s	Connecting systems	Systems started talking to each other automatically
No-Code Automation	2011+	Connecting apps visually	Zapier, IFTTT — anyone could automate without code
AI-Enhanced Automation	2020–2023	Smart task handling	AI doing specific tasks inside workflows (classify, write, analyze)
AI Agentic Era	2023–Now	Full autonomous decision-making	AI controlling entire workflows, using tools, reasoning, acting
Physical AI / Robots	Coming	Real world automation	AI-powered robots handling physical tasks — the cycle comes full circle

The Big Picture Line:

> *"Humans first automated with their hands — making tools. Then with nature — water and wind. Then with machines. Then with software. Now with AI. Every step was about removing human effort from repetitive work. n8n sits right at the AI automation layer — and that's exactly why this is the most exciting time to learn it."*

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4. 3 Types of Workflows in n8n

> *This is the framework for the entire course. Every project you build will fall into one of these.*

Overview Table

Type	Who controls the flow	AI involved	Real world feel
Standard Workflow	You — every step defined	❌ None	Factory conveyor belt
AI-Enhanced Workflow	You — but AI handles specific tasks	✅ AI does tasks	Conveyor belt with a smart worker at one station
AI Agentic Workflow	AI — decides what happens next	✅ AI is the brain	A smart worker who runs the whole factory floor

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Type 1 — Standard Workflow

What it is: Every step is defined by you. Same path every single time. No decisions, no AI, just pure automation.

Characteristics:

- Predictable and reliable
- Breaks if something unexpected happens
- You are the brain — the workflow just executes

Scenario:

> Every day at 9am, fetch yesterday's leads, count them by date, send the total to Slack. Same steps, every day, no thinking needed. That's a Standard Workflow.

Your course projects using this: Lead Capture, Lead Report → Slack

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Type 2 — AI-Enhanced Workflow

What it is: The workflow path is still yours, but AI handles specific tasks inside it — writing, classifying, analyzing, scoring.

Characteristics:

- You still control the overall flow
- AI is a powerful node inside, not the decision maker
- Handles unstructured data (text, responses, opinions)

Scenario:

> Lead submits a form with their answers. Your workflow is fixed — receive → process → send email. But inside the process step, AI reads their answers and *writes* a personalized email for each person. You couldn't do that with just logic and conditions.

The key difference from Standard:

> Standard uses `IF score > 7 THEN send email A`. AI-Enhanced says *"read their answers and write them something relevant"* — AI does the judgment.

Your course project using this: Lead Qualifying + Personalized Email + Mental Health Report

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Type 3 — AI Agentic Workflow

What it is: AI is no longer just doing a task — it's deciding the entire flow. What runs next, what tool to use, whether to ask a follow-up question, whether to trigger another workflow — all decided by AI.

Characteristics:

- Non-linear — path changes based on context
- AI uses tools (search knowledge base, fetch data, trigger workflows)
- Has memory — remembers previous messages in a conversation
- Can loop, self-correct, and plan across multiple steps

Inside this type, two patterns:

Pattern	What it means
Single Agent	One AI running the full agentic workflow
Multi-Agent	Multiple AIs each with a role, collaborating — *teased at end of course*

Scenario:

> User opens chatbot on your website and asks a question. AI decides — should I search the knowledge base? Should I ask a clarifying question? Should I trigger the lead capture workflow? It doesn't follow a fixed path. It reasons and acts. That's Agentic.

The key difference from AI-Enhanced:

> In AI-Enhanced, you decide the flow and AI does tasks. In Agentic, AI decides the flow AND does the tasks.

Your course project using this: Trained Website Chatbot

How the 3 Types Map to Your 4 Projects

Project	Type	What makes it that type
Lead Capture	Standard Workflow	Fixed steps, no AI, Webhook → Filter → Store
Lead Report → Slack	Standard Workflow	Fixed schedule, fixed path, Cron → Count → Send
Lead Qualify + Email	AI-Enhanced Workflow	Your flow, but AI reads answers and writes personalized content
Website Chatbot	AI Agentic Workflow	AI decides what to do next at every step, uses memory and tools

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5. n8n Self-Host vs n8n.io

	Self-Hosted	n8n.io Cloud
Cost	Server cost only (~$5-10/month VPS)	Subscription plan
Setup time	Manual — Docker or VPS	Ready in 2 minutes
Data privacy	100% yours — nothing leaves your server	Stored on their servers
Execution limits	Unlimited	Plan dependent
Maintenance	You handle updates	They handle everything
Custom nodes	✅ Full	⚠️ Limited

Best for	Businesses, sensitive data (health, leads)	Beginners, learning, prototyping

Scenario that makes the choice clear:

In this course, Project 3 handles mental health data. Project 1 handles business leads. Both are sensitive. If you self-host, that data never leaves your machine. If you use n8n.io, it sits on their servers. For learning — n8n.io is perfectly fine. For production with real users — self-host.

n8n (automation platform): https://n8n.io

Hostinger n8n Self-Hosting (20% off with my link): Open Now, More **Affordable Plan**